

Research Plan for Uncertainty-Aware INR Reconstruction of Multi-Beam ISR Plasma Fields

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Abstract

This document consolidates the current state of the implicit neural representation (INR) radar project, the synthetic-data results obtained so far, the proposal commitments, and a recommended research plan for the next stage. The central objective is to reconstruct continuous ionospheric plasma fields from sparse multi-beam incoherent scatter radar (ISR) products while providing explicit reliability maps. The proposed path remains deliberately outside a full physics-informed neural network (PINN): physical knowledge enters as uncertainty-aware likelihoods, anisotropic soft priors, geometry-aware support diagnostics, integration-time-aware synthetic tests, and calibrated reliability products. A key addition in this version is that integration time must be treated as a first-order experimental axis. For fast plasma motion, one-minute, two-minute, five-minute, and ten-minute products are not merely different cadences; they represent different time-averaged measurements of a moving field and can therefore contain motion blur and stationarity errors.

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1 Executive Summary

The project should be framed as an uncertainty-aware continuous interpolation and reconstruction tool for ISR-derived plasma parameters. The tool should produce two coupled outputs:

1. a reconstructed field, initially $\log_{10} N_e(x, y, t)$ and later $\log_{10} N_e(x, y, z, t)$, $T_e(x, y, z, t)$, and $T_i(x, y, z, t)$;
2. a reliability layer that tells users where the reconstruction is data constrained, weakly constrained, or prior dominated.

The work so far supports the following conclusions.

1. A SIREN-like INR is a good baseline because it provides a differentiable continuous field and meaningful derivatives [21].
2. A pure data-only INR can fit sparse observations but may produce unsupported structure and derivative artifacts between beams.
3. Moderate curvature regularization improved the synthetic reconstruction in the first controlled sweep, while stronger regularization was not uniformly beneficial.
4. The current four-case regularization sweep is a pilot, not final evidence. A defensible study requires a structured loss-weight protocol, multiple velocities, multiple integration times, multiple beam counts, and withheld-beam/time validation.
5. Integration time must be included. At 2 km s^{-1} , a plasma feature moves 120 km during a one-minute integration, 240 km during two minutes, 600 km during five minutes, and 1200 km during ten minutes. These are not small perturbations for the current synthetic domains.
6. Eleven-beam experiments should not be discarded. They are likely too sparse for uniformly accurate reconstruction, but they are valuable stress tests for reliability maps. A successful method should warn the user when the eleven-beam reconstruction is not trustworthy.

The recommended next milestone is therefore an integration-aware synthetic benchmark:

1. beam counts: 42, 23, and 11;
2. velocities: 0, 0.36, 0.75, 1.0, 1.5, 2.0, and 3.0 km s^{-1} ;
3. integration times: 1, 2, 5, and 10 minutes;
4. model families: baseline INR in (x, y, t) , velocity-aware co-moving INR, and later full 4D INR;

5. reliability methods: MC dropout plus geometry/support features, and conformal prediction;
6. validation: dense synthetic truth, withheld beams, withheld times, regional metrics, and interval coverage.

2 Project Contract from the Proposal

The UAI proposal frames the project as an uncertainty-aware reconstruction method for 4D ionospheric structure from sparse multi-beam radar data [1]. The core commitments are:

1. fit continuous neural fields for ISR-derived parameters such as N_e , T_e , and T_i ;
2. use ISR-reported errors as heteroscedastic measurement noise;
3. add two interpretable soft priors: temporal continuity and anisotropic smoothness;
4. deliver reliability products that include an observability proxy;
5. validate using withheld-beam and withheld-time tests;
6. benchmark against interpolation and smoothing baselines.

One update is recommended. The proposal mentions ensemble-style uncertainty in some places, but the current plan should not use ensembles as a main method. The uncertainty branch should be:

1. MC dropout plus geometry/support features [29, 30];
2. conformal prediction, especially normalized or group-conditioned conformal intervals [31, 32].

The revised project claim should be:

We reconstruct continuous ISR-derived plasma fields and provide calibrated reliability diagnostics that distinguish data-constrained interpolation from prior-dominated extrapolation.

3 Current Methodological Progress

3.1 Data Representation

The current real-data branch uses sparse radar samples as coordinate-value pairs,

$$\mathcal{D} = \{(x_i, y_i, t_i, u_i)\}_{i=1}^N, \quad u_i = \log_{10} N_{ei}, \quad (1)$$

where (x, y) are horizontal coordinates, t is time, and u is log electron density. Coordinates and targets are normalized during training. The model is a coordinate-based neural field,

$$f_\theta(x, y, t) \approx \log_{10} N_e(x, y, t). \quad (2)$$

The current implementation is not yet full (x, y, z, t) . It uses altitude-filtered slices, which is appropriate at this stage because it keeps the validation problem interpretable.

3.2 Baseline Architecture

The current baseline is a sine-activated MLP inspired by SIREN:

$$h_\ell = \sin(\omega_\ell(W_\ell h_{\ell-1} + b_\ell)). \quad (3)$$

SIREN is attractive because it represents continuous fields and derivatives well [21]. This matters because the project evaluates first and second derivatives for regularization and diagnostics. A ReLU network would be poorly matched to second-derivative curvature penalties because its second derivatives are zero almost everywhere and undefined at kinks.

3.3 Current Loss

The current training objective is:

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda_{xy}\mathcal{L}_{xy} + \lambda_t\mathcal{L}_t. \quad (4)$$

The data loss is currently MSE or synthetic MSE:

$$\mathcal{L}_{\text{data}} = \frac{1}{N} \sum_i (f_\theta(x_i, y_i, t_i) - u_i)^2. \quad (5)$$

The spatial term is a horizontal Hessian penalty:

$$\mathcal{L}_{xy} = \mathbb{E}_{(x,y,t) \sim \Omega} [f_{xx}^2 + 2f_{xy}^2 + f_{yy}^2]. \quad (6)$$

This is not the same as a Laplacian penalty $(f_{xx} + f_{yy})^2$. It penalizes curvature in all horizontal directions, including mixed curvature.

The temporal term is:

$$\mathcal{L}_t = \mathbb{E}_{(x,y,t) \sim \Omega} [f_{tt}^2]. \quad (7)$$

This allows linear temporal trends but discourages rapid temporal bending.

3.4 Synthetic Validation Infrastructure

The synthetic branch was created because real radar data do not provide dense truth. The synthetic case allows the same INR training workflow to be tested against a known analytic plasma field. The completed progress includes:

1. synthetic moving-patch generator;
2. sparse radar-like observation geometries;
3. 42-beam and 23-beam sampling patterns;
4. data-only and regularized training cases;
5. derivative diagnostics even when regularization is disabled;
6. dense reconstruction analysis using the corrected reconstruction/error script;
7. regional error metrics: full domain, interior, near-observation, and high-gradient regions;
8. comparison plots and summary tables.

3.5 First Synthetic Sweep Results

The first controlled sweep used a moving Gaussian-like density patch at approximately 360 m s^{-1} . The key cases were:

1. data-only;
2. moderate regularization: $\rho_{xy} = 0.30$, $\rho_t = 0.30$;
3. stronger spatial regularization: $\rho_{xy} = 0.70$, $\rho_t = 0.30$;
4. stronger spatial and temporal regularization: $\rho_{xy} = 0.70$, $\rho_t = 0.70$.

The main finding was that moderate regularization was best overall in the first benchmark. It reduced artifacts and improved gradient recovery without over-smoothing as strongly as the higher regularization cases. The data-only run was competitive for scalar density in some high-gradient regions, but derivative recovery and artifact control were weaker. This supports the project hypothesis that priors are useful, but it does not yet prove that any particular weight is optimal.

4 Why Integration Time Must Be Included

4.1 Cadence and Integration Time Are Different Concepts

ISR products commonly appear at integration times such as 1, 2, 5, and 10 minutes. It is tempting to treat these products as point samples at their timestamps. That is only approximately true when the plasma is slowly varying over the integration interval. For fast plasma flow, a radar product is better interpreted as a time-averaged measurement over a moving plasma field.

Two time scales must be separated:

1. sampling interval or cadence, Δt_{samp} : the time between successive records;
2. integration duration, T_{int} : the time over which the measurement is accumulated.

Often these are similar in processed products, but conceptually they have different effects. Cadence controls the displacement between frames. Integration duration controls motion blur within each frame.

4.2 Forward Model for Integrated Observations

An instantaneous synthetic observation would be:

$$y_i = u(\mathbf{r}_i, t_i) + \epsilon_i, \quad (8)$$

where $\mathbf{r}_i = (x_i, y_i, z_i)$ or (x_i, y_i) .

An integration-aware observation should instead be:

$$y_i = \frac{1}{T_{\text{int}}} \int_{t_i - T_{\text{int}}/2}^{t_i + T_{\text{int}}/2} u(\mathbf{r}_i, \tau) d\tau + \epsilon_i. \quad (9)$$

If the plasma field is a moving structure,

$$u(\mathbf{r}, t) = u_0(\mathbf{r} - \mathbf{v}t), \quad (10)$$

then Eq. 9 is a convolution of the instantaneous structure with a line segment in the direction of motion. The measurement is not just noisy; it is blurred along the flow direction.

4.3 Motion Blur for a Gaussian Patch

For a Gaussian patch with horizontal covariance Σ_0 moving at velocity $\mathbf{v}$, a uniform integration interval approximately adds covariance along the direction of motion:

$$\Sigma_{\text{eff}} \approx \Sigma_0 + \frac{T_{\text{int}}^2}{12} \mathbf{v} \mathbf{v}^\top. \quad (11)$$

This is the covariance of a Gaussian convolved with a uniform line segment. The important consequence is that motion blur is anisotropic and grows quadratically with integration time.

4.4 Dimensionless Severity Numbers

The synthetic benchmark should track dimensionless numbers rather than only raw velocities. Let ℓ be a characteristic spatial scale, such as patch width, beam spacing, or the smallest feature scale we expect to reconstruct. Define:

$$C_{\text{samp}} = \frac{\|\mathbf{v}\| \Delta t_{\text{samp}}}{\ell}, \quad (12)$$

$$C_{\text{int}} = \frac{\|\mathbf{v}\| T_{\text{int}}}{\ell}. \quad (13)$$

Interpretation:

1. $C_{\text{samp}} \ll 1$: the feature moves little between records;
2. $C_{\text{samp}} \sim 1$: the feature moves by about one feature width between records;
3. $C_{\text{samp}} > 1$: temporal interpolation becomes a tracking problem;
4. $C_{\text{int}} \ll 1$: the integrated product is close to instantaneous;
5. $C_{\text{int}} \sim 1$: the measurement is visibly motion blurred;
6. $C_{\text{int}} > 1$: a point-sample interpretation is physically questionable.

4.5 Displacements for Candidate Velocities and Integration Times

Table 1 shows how far a structure moves during common ISR integration intervals. These values strongly argue for including integration time in the synthetic benchmark.

Table 1: Horizontal displacement during integration for candidate synthetic velocities.

Velocity (km s ⁻¹)	1 min km	2 min km	5 min km	10 min km
0.00	0	0	0	0
0.36	21.6	43.2	108	216
0.75	45	90	225	450
1.00	60	120	300	600
1.50	90	180	450	900
2.00	120	240	600	1200

Velocity (km s ⁻¹)	1 min km	2 min km	5 min km	10 min km
3.00	180	360	900	1800

At 2 km s⁻¹, a 5-minute product averages over roughly 600 km of motion. For many radar volumes this is not a detail; it changes the inverse problem. A 10-minute product at active-time velocities may be excellent for reducing stochastic fit uncertainty but poor for recovering instantaneous morphology.

4.6 Noise-Blur Tradeoff

Longer integration generally reduces stochastic measurement noise. A simple first-order rule is:

$$\sigma_{\text{stat}}(T_{\text{int}}) \propto \frac{1}{\sqrt{T_{\text{int}}}}. \quad (14)$$

But longer integration increases motion-induced model mismatch. A local first-order approximation is:

$$\sigma_{\text{motion}}^2 \approx \frac{T_{\text{int}}^2}{12} \left(\mathbf{v}^\top \nabla u \right)^2. \quad (15)$$

Therefore an effective uncertainty for fast-moving structures should include both:

$$\sigma_{\text{eff}}^2 = \sigma_{\text{fit}}^2 + \sigma_{\text{floor}}^2 + \sigma_{\text{motion}}^2 + \sigma_{\text{systematic}}^2. \quad (16)$$

This expression should not be presented as final ISR theory yet. It is a practical synthetic and diagnostic model that prevents long-integration products from being treated as overconfident point measurements during active times.

5 Recommended Synthetic Velocity Suite

5.1 Velocity Values

The next synthetic suite should include:

$$\|\mathbf{v}\| \in \{0, 0.36, 0.75, 1.0, 1.5, 2.0, 3.0\} \text{ km s}^{-1}. \quad (17)$$

Interpretation:

Velocity	Purpose
0	static control
0.36 km s ⁻¹	quiet/moderate baseline, current case
0.75 km s ⁻¹	active but usually still trackable at short cadence
1.0 km s ⁻¹	strong convection
1.5 km s ⁻¹	active/storm-time stress
2.0 km s ⁻¹	severe active-time case
3.0 km s ⁻¹	out-of-distribution stress test

These values are also physically interpretable from the $\mathbf{E} \times \mathbf{B}$ drift scale. For a high-latitude magnetic field of order $B \sim 5 \times 10^{-5}$ T,

$$v_E = \frac{E}{B}. \quad (18)$$

Thus $E = 50$ mV m⁻¹ corresponds to approximately 1 km s⁻¹, and $E = 100$ mV m⁻¹ corresponds to approximately 2 km s⁻¹. Such values are not quiet-time values, but they are appropriate stress cases for active high-latitude convection [19, 17, 18].

5.2 Velocity Directions

Do not use only one direction. Use a small set of directions:

$$\phi \in \{0^\circ, 45^\circ, 90^\circ, 135^\circ\}, \quad (19)$$

or align directions relative to the radar beam fan:

1. along the densest beam fan;
2. across the beam fan;
3. diagonal to the beam geometry;
4. approximately field-aligned projection if magnetic coordinates are introduced.

The same speed can be easy or hard depending on whether the structure moves along well-sampled directions or through gaps between beams.

6 Beam Count Strategy

The current synthetic work includes 42-beam and 23-beam cases. An 11-beam case should be added, but it should be interpreted correctly.

Beam count	Role	Expected outcome
42	optimistic/method-development case	should reconstruct moderate motion well
23	realistic sparse benchmark	main scientific benchmark
11	stress/failure case	should expose low-reliability regions

The 11-beam case may indeed be too sparse for strong reconstruction. That is not a reason to omit it. The final tool must be able to tell a user when an interpolated field is not trustworthy. If an 11-beam high-speed case produces a visually smooth reconstruction but the reliability layer correctly marks large regions as unreliable, that is a project success.

7 Synthetic Experiment Matrix

7.1 Stage 1: Manageable Core Matrix

A full factorial sweep over velocity, direction, integration time, beam count, architecture, and loss weights would become too large. The first serious matrix should be:

1. beam counts: 42, 23, 11;

2. velocities: 0.36, 1.0, 2.0 km s⁻¹;
3. integration times: 1, 2, 5, 10 minutes;
4. direction: one standard direction, preferably the current direction;
5. regularization: data-only, moderate baseline, and selected stronger case.

This is already:

$$3 \times 3 \times 4 \times 3 = 108$$

training conditions if all are run. If runtime is high, reduce Stage 1 to:

$$\text{beams} \in \{42, 23, 11\}, \quad v \in \{0.36, 1.0, 2.0\}, \quad T_{\text{int}} \in \{1, 5, 10\},$$

with only the moderate regularization setting and a data-only control for selected cases.

7.2 Stage 2: Directional Stress

After Stage 1, fix the most informative speeds, probably 1.0 and 2.0 km s⁻¹, and test several directions. This tells us whether performance is controlled by speed alone or by speed relative to beam geometry.

7.3 Stage 3: Velocity-Aware Models

Compare:

1. standard INR:

$$f_{\theta}(x, y, t);$$

2. co-moving coordinate INR:

$$f_{\theta}(x - u_x t, y - u_y t, t);$$

3. weak advection diagnostic, not yet a training loss:

$$r_{\text{adv}} = f_t + u_x f_x + u_y f_y.$$

The co-moving coordinate version is recommended before an advection residual loss because it is still an interpolation representation rather than a PDE-constrained PINN. If an advection residual is later used, it should be presented as a diagnostic or a soft transport prior, not as a full physical model.

8 Using Line-of-Sight Velocities

Radar products include line-of-sight velocity. These should be used, but carefully.

8.1 Synthetic Velocity Generation

For synthetic data, define a true horizontal velocity:

$$\mathbf{v}_h = (u_x, u_y). \quad (20)$$

For beam unit vector $\hat{\mathbf{b}}_j$, the synthetic line-of-sight velocity is:

$$v_{\text{los}}^{(j)} = \mathbf{v}_h \cdot \hat{\mathbf{b}}_h^{(j)} + \epsilon_v, \quad (21)$$

where $\hat{\mathbf{b}}_h^{(j)}$ is the horizontal projection of the beam direction. This allows the synthetic benchmark to include realistic velocity observations without pretending that every beam directly measures the full horizontal vector.

8.2 First Real-Data Use

For real data, the first use of v_{los} should be conservative:

1. use v_{los} as a speed/activity proxy;
2. use it to flag intervals where long integration times may be motion blurred;
3. use it as an input feature or reliability feature;
4. use it to estimate σ_{motion} ;
5. avoid claiming a full vector velocity field from monostatic LOS data without additional assumptions.

Later, a smoothed horizontal velocity field can be estimated from multiple beams or external context, but that should be a separate methodological step.

9 ISR Uncertainty and Error Modeling

9.1 Formal Fit Errors

The ISR error document emphasizes that reported variables such as dN_e , dT_e , dT_i , and dV_i are formal fitting uncertainties, not total truth errors [2]. They are derived from the nonlinear spectral fitting problem and reflect the local posterior covariance under the assumed forward model [8, 9, 14, 15].

For a transformed target $u = \log_{10} N_e$, the uncertainty should be transformed as:

$$\sigma_u \approx \frac{\sigma_{N_e}}{\ln(10)N_e}. \quad (22)$$

9.2 Heteroscedastic Data Loss

The data loss should become:

$$\mathcal{L}_{\text{data},\sigma} = \frac{1}{N} \sum_{i=1}^N \frac{(f_{\theta}(\mathbf{x}_i, t_i) - u_i)^2}{\sigma_{\text{eff},i}^2}. \quad (23)$$

The effective uncertainty should include a floor:

$$\sigma_{\text{eff},i} = \max(\sigma_{u,i}, \sigma_{\text{floor}}), \quad (24)$$

or a richer expression:

$$\sigma_{\text{eff},i}^2 = \sigma_{u,i}^2 + \sigma_{\text{floor}}^2 + \sigma_{\text{motion},i}^2 + \sigma_{\text{support},i}^2. \quad (25)$$

9.3 Normalized Residual Diagnostics

Use:

$$z_i = \frac{f_\theta(\mathbf{x}_i, t_i) - u_i}{\sigma_{\text{eff},i}}. \quad (26)$$

Then track:

$$\chi_\nu^2 \approx \frac{1}{N} \sum_i z_i^2. \quad (27)$$

Values near one indicate rough consistency if the error model is appropriate. Values much larger than one suggest under-estimated uncertainty or model mismatch. Values much smaller than one suggest over-estimated uncertainty or over-smoothed predictions.

9.4 Systematic Effects Not Captured by Formal Error Bars

Formal fit uncertainties do not fully capture:

1. ion composition assumptions;
2. non-Maxwellian distributions;
3. beam-to-beam calibration bias;
4. range-lag ambiguity and range mixing;
5. velocity shear during integration;
6. low-elevation gain/SNR degradation;
7. I/Q and processing artifacts;
8. temporal nonstationarity during integration.

These should feed the reliability map even when they are not included directly in training.

10 Anisotropy and the Vertical Dimension

10.1 Horizontal and Vertical Directions Should Differ

The vertical coordinate z should not be treated as simply another horizontal coordinate. The ionosphere has strong vertical stratification, layer structure, scale-height behavior, composition transitions, and range ambiguity. Horizontal interpolation across sparse beams and vertical interpolation across range gates have different physical meanings.

For full 4D reconstruction, use:

$$f_\theta(x, y, z, t) = b_\phi(z, t) + r_\theta(x, y, z, t), \quad (28)$$

where b_ϕ is a smooth background vertical profile and r_θ is a residual 4D INR. This decomposition gives the model a physically meaningful vertical baseline while allowing horizontal structure.

10.2 Anisotropic Curvature

The 4D loss should generalize to:

$$\mathcal{L} = \mathcal{L}_{\text{data},\sigma} + \lambda_{xy}\mathcal{L}_{xy} + \lambda_z\mathcal{L}_z + \lambda_t\mathcal{L}_t. \quad (29)$$

The vertical penalty can be:

$$\mathcal{L}_z = \mathbb{E}[f_{zz}^2], \quad (30)$$

but for sharp layers a robust penalty should be tested:

$$\mathcal{L}_z^{\text{Huber}} = \mathbb{E}[\rho_\delta(f_{zz})], \quad (31)$$

where ρ_δ is the Huber loss. Cauchy or total-variation-like penalties are also worth testing because full-profile ISR analysis and Bayesian inversion literature warns that stationary Tikhonov-type smoothing can suppress real sharp structures [2, 20].

10.3 Magnetic Coordinates

For high-latitude data, geographic (x, y, z) may not always be the best coordinate system. Potential future coordinate systems include:

1. geographic/cartesian coordinates;
2. magnetic apex coordinates;
3. coordinates decomposed into directions parallel and perpendicular to $\mathbf{B}$;
4. beam/range/time coordinates for data-ingest diagnostics.

The EISCAT 3D uncertainty literature emphasizes the importance of geometry, viewing angle relative to the magnetic field, and experiment design [16].

11 Architecture Roadmap

11.1 SIREN as the Main Baseline

SIREN remains the correct baseline because:

1. it maps coordinates to continuous values;
2. it can represent smooth fields and derivatives;
3. it works naturally with derivative regularization;
4. frequency scaling acts as a spectral prior.

However, the SIREN frequency parameter ω_0 must be treated as a hyperparameter and not as a universal constant.

11.2 FINER and FINER++

FINER and FINER++ introduce variable-periodic activation functions to tune spectral support more flexibly than standard SIREN [23, 24]. These should be tested if:

1. low-frequency SIREN is too smooth;
2. high-frequency SIREN creates ringing;
3. one fixed ω_0 cannot represent both smooth background and sharp structures.

11.3 K-Planes and Factorized 4D Representations

K-Planes factorizes high-dimensional fields into learned planes and is explicitly designed to move from static to dynamic scenes [25]. It is attractive for full (x, y, z, t) reconstruction because it can make dimension-specific priors easier to implement. For example, temporal smoothness can be applied to time-containing planes and vertical priors can be separated from horizontal ones.

11.4 Hash or Grid Hybrid Representations

Multiresolution hash encodings can drastically accelerate neural field training [26]. They should be considered if training cost becomes limiting. However, they may make derivative priors less clean, so they should not replace SIREN before the uncertainty-aware baseline is stable.

11.5 Radar Neural Fields

Radar-neural-field papers such as RF4D and SpINR are conceptually relevant because they show that radar-like data can be represented by neural fields [27, 28]. However, they are not directly equivalent to ISR plasma-parameter reconstruction. ISR products are fitted plasma parameters with formal uncertainties, not raw FMCW point clouds or automotive radar returns.

12 Loss-Weight Selection Protocol

12.1 Why the Current Sweep Is Not Final

The four-case sweep is scientifically useful but not sufficient as final evidence. It showed that regularization matters and that stronger is not automatically better. A final study requires a structured selection protocol.

12.2 Recommended Weight Grid

Use target ratios rather than raw lambdas:

$$\rho_{xy} = \frac{\lambda_{xy}\mathcal{L}_{xy}}{\mathcal{L}_{\text{data}}}, \quad \rho_t = \frac{\lambda_t\mathcal{L}_t}{\mathcal{L}_{\text{data}}}. \quad (32)$$

Recommended first grid:

$$\rho_{xy}, \rho_t \in \{0, 0.05, 0.1, 0.3, 0.7, 1.0\}. \quad (33)$$

Select weights by validation performance, not training loss. The primary selection metrics should be withheld-beam error, withheld-time error, high-gradient error, and calibration quality.

12.3 Adaptive Loss Balancing

Compare the fixed-ratio method with:

1. L-curve analysis for the data-fit/regularization tradeoff [20];
2. GradNorm-style gradient balancing [33];
3. SoftAdapt-style loss-rate balancing [34];
4. uncertainty weighting for multi-task losses [35];
5. ReLoBRaLo-style balancing, used only as a multi-loss optimizer idea, not as a PINN claim [37];
6. gradient-flow diagnostics inspired by PINN loss pathology studies [36].

The most defensible phrasing is:

We seek the smoothest and most reliable field whose uncertainty-normalized data misfit remains acceptable under withheld-beam and withheld-time validation.

13 Reliability Maps

13.1 Reliability Is a Deliverable, Not a Decoration

The final tool should show both interpolation and reliability. A region marked unreliable is not a failure; it is valuable scientific information. The reliability map should warn users when the model is extrapolating across poorly observed space, long integration time, high motion, low SNR, or high dropout/conformal uncertainty.

13.2 MC Dropout Route

MC dropout provides stochastic forward passes:

$$\{\hat{u}^{(m)}(\mathbf{x}, t)\}_{m=1}^M. \quad (34)$$

From these compute:

$$\mu(\mathbf{x}, t) = \frac{1}{M} \sum_m \hat{u}^{(m)}(\mathbf{x}, t), \quad (35)$$

$$\sigma_{\text{MC}}^2(\mathbf{x}, t) = \frac{1}{M-1} \sum_m \left(\hat{u}^{(m)}(\mathbf{x}, t) - \mu(\mathbf{x}, t) \right)^2. \quad (36)$$

This is an epistemic uncertainty proxy [29, 30].

13.3 Conformal Prediction Route

Conformal prediction wraps a trained model with calibrated intervals. For calibration samples, define a normalized score:

$$s_i = \frac{|u_i - \hat{u}_i|}{a_i}, \quad (37)$$

where a_i is a scale factor. For this project:

$$a_i^2 = \sigma_{\text{ISR},i}^2 + \sigma_{\text{MC},i}^2 + \sigma_{\text{motion},i}^2 + \sigma_{\text{support},i}^2. \quad (38)$$

Let $q_{1-\alpha}$ be the appropriate conformal quantile of the calibration scores. Then:

$$\hat{u}(\mathbf{x}, t) \pm q_{1-\alpha} a(\mathbf{x}, t) \tag{39}$$

is a calibrated prediction interval under the assumptions of the conformal split [31, 32].

13.4 Group-Conditioned Reliability

A single global interval may be misleading. Use bins or groups:

1. altitude range;
2. support distance;
3. beam elevation;
4. SNR or reported σ ;
5. integration time;
6. velocity/motion-blur class;
7. interior vs boundary regions.

This is similar in spirit to Mondrian or group-conditioned conformal prediction, and it helps ensure that high-confidence regions and weak-support regions are not averaged into one misleading calibration score.

13.5 Reliability Map Components

The reliability map should combine:

1. ISR aleatoric uncertainty;
2. MC dropout dispersion or conformal interval width;
3. distance to nearest measurement;
4. beam density/coverage;
5. beam-angle diversity;
6. integration-time motion blur;
7. window-overlap disagreement;
8. withheld-validation residual patterns;
9. quality flags and systematic-risk indicators.

14 Validation Design

14.1 Synthetic Validation

Synthetic validation should include:

1. dense truth error;
2. gradient error;
3. high-gradient region error;
4. boundary vs interior error;
5. near-observation vs far-support error;
6. motion-speed and integration-time stratification;
7. beam-count stratification.

14.2 Real-Data Validation

Real-data validation should use:

1. random withheld points for debugging;
2. withheld range gates;
3. withheld altitude bands;
4. withheld time blocks;
5. withheld beams;
6. combined withheld beam/time stress tests.

14.3 Metrics

Report:

1. RMSE, MAE, bias, and 95th percentile absolute error;
2. normalized residuals and reduced chi-square-like metrics;
3. empirical coverage of nominal 50%, 80%, and 90% intervals;
4. interval width;
5. coverage by altitude;
6. coverage by support-distance bin;
7. coverage by beam/elevation/SNR bin;
8. coverage by integration time and velocity class.

15 Recommended Final Tool

The final deliverable should be an interactive analysis tool, not only a training script. It should show:

1. reconstructed $\log_{10} N_e$ slices;
2. later T_e and T_i slices;
3. observed beam/range samples overlaid;
4. reliability map;
5. interval-width map;
6. support/observability map;
7. integration-time or motion-blur warning layer;
8. optional mask that hides low-reliability regions;
9. calibration summary, e.g. “nominal 90% interval achieved 87% coverage on withheld beams.”

The most important user-interface principle is that unreliable regions should remain visible but clearly marked. The scientific value is not only in showing the interpolation, but in showing where the interpolation should not be trusted.

16 Recommended Near-Term Tasks

16.1 Task 1: Add Integration-Aware Synthetic Generator

Modify the synthetic observation generator so each observation can be either instantaneous or integrated:

$$y_i^{\text{inst}} = u(\mathbf{r}_i, t_i), \quad y_i^{\text{int}} = \frac{1}{T_{\text{int}}} \int u(\mathbf{r}_i, \tau) d\tau. \quad (40)$$

Numerically approximate the integral by sub-sampling within the integration window.

16.2 Task 2: Add Velocity and Integration Metadata

Every synthetic output directory should record:

1. true velocity vector;
2. speed;
3. direction;
4. integration time;
5. cadence;
6. beam count;
7. feature width;
8. C_{samp} and C_{int} .

16.3 Task 3: Add 11-Beam Stress Geometry

Use 11 beams as a reliability stress test. Do not expect uniform accuracy. Expect the reliability map to degrade in unsampled regions.

16.4 Task 4: Compare Instantaneous-Truth and Integrated-Truth Metrics

For integrated products, evaluate against two truths:

1. instantaneous truth at the nominal timestamp;
2. integrated truth over the same time window.

If the model is trained on integrated observations, it may be unfair to demand exact instantaneous truth recovery unless a deblurring or motion-compensation model is included.

16.5 Task 5: Add Velocity-Aware Coordinates

Test:

$$(x, y, t) \quad \text{vs.} \quad (x - u_x t, y - u_y t, t). \quad (41)$$

This is the lowest-complexity way to see whether known or estimated velocity can improve fast-moving reconstruction without turning the project into a PINN.

16.6 Task 6: Add Reliability Prototype

Start with:

$$R(\mathbf{x}, t) = g(d_{\text{nearest}}, \sigma_{\text{ISR}}, \sigma_{\text{MC}}, T_{\text{int}}, \|\mathbf{v}\|, \text{beam density}), \quad (42)$$

where R is a reliability score and g is initially a transparent hand-designed function. Later replace or calibrate this with conformal intervals.

17 Conclusion

The project is no longer simply testing whether a SIREN can draw a plausible plasma-density image. It is becoming a statistically and physically informed reconstruction framework for sparse ISR products. The key scientific question is not merely whether the INR can interpolate. The key question is:

Can the INR interpolate sparse ISR plasma fields while honestly indicating where the result is data constrained and where it is not?

The next stage should make velocity, integration time, beam sparsity, uncertainty, and reliability central experimental axes. In particular, integration time is mandatory. At active-time velocities, one-minute, two-minute, five-minute, and ten-minute products correspond to fundamentally different inverse problems. A reliable reconstruction tool must know this, expose it, and warn the user when long integration, sparse beams, or fast motion make the interpolation unreliable.

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